Phase 1 – Problem Statements and Solutions
(Detecting Outliers in Financial Transactions)

1) Unusually High Transaction Amounts

Problem

Some transactions have abnormally high values that may indicate fraud or errors.

Expected Solution

Detect high-value outlier transactions for investigation.

How (Process of Solution)

- Analyze transaction amounts.
- Use IQR/Z-Score to identify outliers.
- Flag abnormal transactions.
- Review flagged transactions.

Analysis

Transaction Amount Outlier Analysis (Transaction Amount vs Outlier Status)

Data Required

- Transaction ID
- Transaction Amount
- Transaction Date
- Customer ID
- Transaction Type

2) Abnormal Customer Transaction Behaviour

Problem

Certain customers perform transactions that differ significantly from their normal behaviour.

Expected Solution

Identify customers with unusual transaction patterns.

How (Process of Solution)

- Analyze customer transaction history.
- Compare current vs historical behaviour.
- Flag abnormal activity.
- Investigate suspicious accounts.

Analysis

Customer-wise Transaction Analysis (Customer vs Transaction Amount)

Data Required

- Customer ID
- Transaction Amount
- Transaction Date
- Transaction Frequency
- Account Type

3) Unusual Merchant Activity

Problem

Some merchants receive unusually high or low transaction volumes.

Expected Solution

Identify merchants with abnormal transaction behaviour.

How (Process of Solution)

- Analyze merchant transactions.
- Compare merchant averages.
- Detect outliers.
- Recommend further review.

Analysis

Merchant-wise Transaction Analysis (Merchant vs Transaction Volume)

Data Required

- Merchant ID
- Merchant Name
- Transaction Amount
- Transaction Count
- Location

4) Suspicious Time-based Transactions

Problem

Transactions occurring at unusual times may indicate fraudulent activity.

Expected Solution

Identify transactions occurring outside normal business patterns.

How (Process of Solution)

- Analyze transaction timestamps.
- Compare with normal activity.
- Detect time-based anomalies.
- Generate alerts.

Analysis

Time-based Transaction Analysis (Time vs Transaction Count)

Data Required

- Transaction ID
- Transaction Time
- Customer ID
- Transaction Amount
- Location

5) Lack of Outlier Monitoring Dashboard

Problem

Organizations lack real-time visibility of abnormal financial transactions.

Expected Solution

Develop a dashboard for monitoring outliers and risk indicators.

How (Process of Solution)

- Create KPIs.
- Visualize outlier trends.
- Build interactive dashboard.
- Support decision-making.

Analysis

Outlier Monitoring Dashboard (KPIs & Risk Summary)

Data Required

- Transaction ID
- Amount
- Date
- Customer
- Merchant
- Risk Flag